

Vinay Venkatesh

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Machine Learning/Statistics and Artificial Intelligence

June 2027

- **Relevant Coursework:** Machine Learning, Computer Vision, Deep Learning, Data Structures and Algorithms, Probability, Data Science, Functional Programming, Linear Algebra, Statistical Visualization
- **Activities:** Data Science Club, Artificial Intelligence Club, Asian Student Association, Phi Delta Theta Fraternity

EXPERIENCE

Software Engineer Intern

May 2026 – Aug. 2026

LinkedIn

Mountain View, CA

- AI Infrastructure - Incoming Summer 2026

AI Integration Engineer Intern

Jan. 2026 – Apr. 2026

Wolfe, LLC

Pittsburgh, PA

- Reduced campaign production time by 98% (4 hr to 5 min) using AWS pipeline (Lambda, Step Functions, Bedrock) to generate marketing content across 5 brands and 3 tiers
- Integrated Triple Whale analytics for data-driven feedback loops, using post-campaign metrics to optimize AI prompts and targeting
- Designed YAML configuration system enabling brand voice, AI prompts, and template modifications without code changes
- Built Flask dashboard on AWS App Runner with campaign launching, pipeline monitoring, and 1-click deployments

Software Engineer Intern

June 2025 – Aug. 2025

Payactiv

Milpitas, CA

- Developed 30+ RESTful APIs for an XMPP-based chat platform supporting 1M+ users
- Designed and updated PostgreSQL schemas, reducing query latency by 15%
- Wrote automated tests with xUnit and Postman, improving API reliability and reducing regressions
- Delivered QBR-ready features two weeks early using Jira within Agile/Scrum workflows

Machine Learning Engineer Intern

Feb. 2023 – Aug. 2024

San Jose State University

San Jose, CA

- Optimized ML pipeline linking 3D facial geometry and DNA, achieving 0.73 AUROC in face-DNA verification
- Compressed 120K+ vertex meshes into 16-D embeddings using SpiralNet++ for efficient geometric learning
- Improved verification accuracy by 18% using 41 biologically tested SNPs instead of PCA on 3.8M genomic markers

Software Engineer Intern

July 2022 – Sep. 2022

GenerAI

Fargo, ND

- Built a web and mobile app using React Native and ASP.NET Core connecting 200+ creators with 50+ brands
- Collaborated with COO and 3-person engineering team to launch MVP within 8 weeks

PROJECTS

ArmVision | Python, Computer Vision, Tensorflow, Research

IEEE Publication Link

- Developed a real-time firearm detection system using computer vision
- Trained a YOLOv5 model on a curated RoboFlow dataset, achieving 93% peak accuracy
- Evaluated model performance across precision, recall, and latency benchmarks
- Published and presented findings at an IEEE conference

TradeSim | Node.js, Express.js, RESTful APIs, Fintech, Full-Stack Development

GitHub Link

- Built a responsive trading dashboard displaying 6 real-time KPIs (volume, open orders, executed trades, etc.)
- Implemented an order matching engine with partial fills, price-priority execution, and multi-symbol order books
- Developed 9 RESTful API endpoints for order submission, matching, event streaming, and more

Math Performance Analysis & Visualization | Next.js, TypeScript, D3.js, Recharts

GitHub Link

- Designed and developed web dashboard to explore factors affecting Portuguese students' math performance
- Created 10+ interactive visualizations revealing relationships between academic habits, demographics, and results

SKILLS

Languages: Python, Java, TypeScript, C, C++, Go, Swift, C#, JavaScript, SQL (Postgres), HTML/CSS, R

Frameworks & Libraries: React, PyTorch, Angular, Node.js, Tensorflow, pandas, NumPy, Matplotlib

Tools: Claude Code, GitHub, Docker, Kubernetes, Azure, Jira, Postman, Cursor, Google Colab, Linux, Kafka

Methodologies: Retrieval Augmented Generation, Time-Series Forecasting, Regression, GANs, PCA, Agile, Scrum